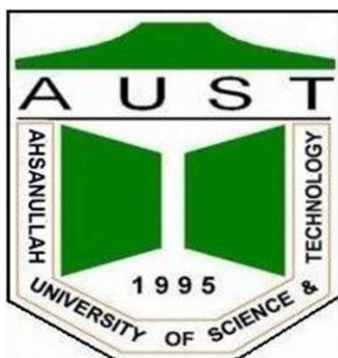




Ahsanullah University of Science & Technology

Department of Mechanical & Production Engineering

Instrument & Measurement Sessional

ME-3110

Project Report

On

“WATER QUALITY MONITORING SYSTEM”

SUBMITTED BY

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26 September, 2024

Ms. Most. Naznin Nahar

Lecturer,

Dept of Mechanical and Production Engineering,

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Subject: Submission of ME-3110 Report on “Water Quality Monitoring System ”.

Mam,

As a part of our Instrument and Measurement (ME 3110) course, we hereby submit the report based on our project and reflecting all the learnings we have gained during this course. As our given theme was making a monitoring system so our chosen project was “Water Quality Monitoring System”

We may, therefore, hope that you would accept the timely submission and oblige thereby. For any kind of explanation and clarification about any topic of the report, we will remain on standby as and when required.

Sincerely yours,

Group-B7

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Acknowledgement

Our utmost gratitude goes to Mr. Badhan Saha, Mr. Syed Shaheer Uddin Ahmed and our supervisor Ms. Most Naznin Nahar, who has guided us throughout the whole course. Their continuous support, motivation, supervision and enthusiasm has resulted into the initiation and execution of this project.

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Introduction

A water quality monitoring system measures various parameters like pH, temperature, and pollutants in water to ensure safety for drinking, agriculture, and the environment. It uses sensors, data loggers, and remote monitoring for real-time analysis. The system is vital for environmental protection, public health, and industrial water management.

Objective

1. Ensure water is safe for drinking and human use.
2. Detect and control pollution sources.
3. Monitor environmental health and biodiversity in water bodies.
4. Ensure compliance with regulatory standards.
5. Provide real-time data for decision-making and emergency response.
6. Support sustainable water management for agriculture and industry.
7. Enable predictive maintenance to prevent system failures or contamination.

Features

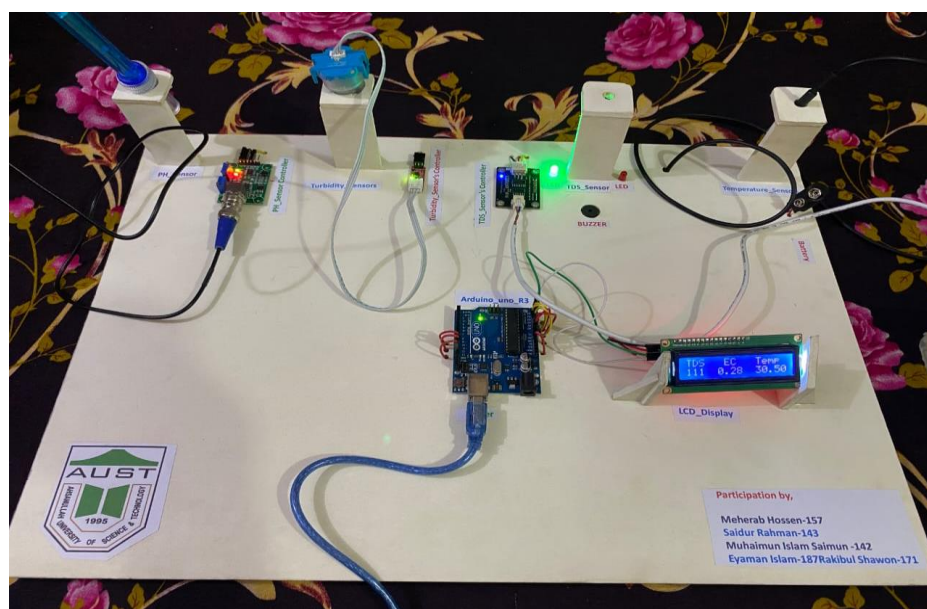
1. Real-time monitoring of water parameters (pH, turbidity, dissolved oxygen, etc.).
2. Wireless data transmission for remote access.
3. Automated data collection using sensors and probes.
4. Data logging and storage for trend analysis.
5. Cloud integration for centralized access and management.
6. Customizable alerts for abnormal conditions.
7. User-friendly interface for visualization and data interpretation.

Description

Life requires water to sustain biological processes including digestion, waste elimination, and temperature regulation. By offering a variety of species' habitats, it preserves the health of the environment. Water is essential to agriculture because it keeps cattle alive and irrigates crops, both of which contribute to food security. Water is a resource that many sectors rely on for production. Clean water is also essential for hygiene and sanitation, which helps to stop the spread of disease. In addition to regulating climate and offering recreational possibilities, water bodies have a crucial role in ecosystem health, economic development, and public health, making sustainable water management imperative.

A technological framework known as a water monitoring system is intended to monitor, assess, and control the amount and quality of water in different ecosystems, including lakes, rivers, and municipal supply systems. Typically, it has sensors that are positioned strategically to collect a wealth of data, detecting things like pollutants, turbidity, pH levels, and dissolved oxygen. For real-time monitoring and analysis, the gathered data is saved by a data logger and sent via a communication network to a central server or cloud platform. The data may be easily interpreted thanks to an intuitive software interface that provides visualizations and alarms for unusual findings. Numerous systems are battery- or solar-powered, guaranteeing uninterrupted operation even in isolated locations. Water monitoring systems are used for a variety of purposes, including agriculture optimization, environmental protection, urban management, providing vital data for sustainable water resource management and public health.

Diagram:



COMPONENTS

Arduino Uno: Arduino acts like the heart of the project . Arduino Uno is a popular microcontroller board based on the ATmega328P chip, widely used for electronics projects. It features 14 digital I/O pins, 6 analog inputs, and a USB interface for easy programming. Open-source and beginner-friendly, it's ideal for robotics, home automation, and prototyping



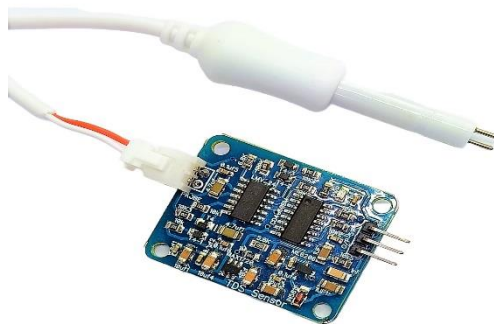
LCD Display: Liquid crystals are used in LCD (Liquid Crystal Display) flat-panel display technology to regulate light, resulting in vivid colors and crisp images. It has a broad viewing angle, good resolution, and is lightweight and thin. It also uses less energy. LCDs are widely used in TVs, monitors, cellphones, and other electronic devices.



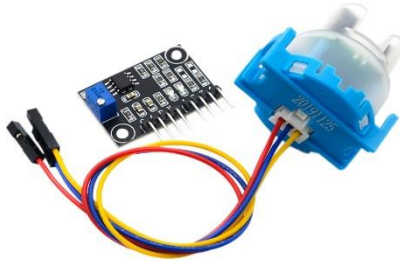
pH Sensor: A pH sensor is essential for many applications, such as monitoring water quality, agriculture, industrial processes, aquaculture, and laboratory studies. It determines the acidity or alkalinity of a solution. It contributes to quality assurance and safety by giving precise pH values in these domains.



Turbidity Sensor: A turbidity sensor detects suspended particles in a liquid by measuring its cloudiness. Light is emitted, scattered light is detected, and the results are reported in NTU (Nephelometric Turbidity Units). These ruggedized sensors are widely utilized in environmental research, wastewater treatment, and water quality monitoring since they offer real-time data.



TDS Sensor: A TDS (Total Dissolved Solids) sensor gauges the concentration of dissolved substances like salts and minerals in water, usually by measuring its conductivity. It displays results in parts per million (ppm) and is commonly used for water quality monitoring, aquaculture, and hydroponics. These sensors offer real-time readings and are designed to be durable for use in different environments.



Battery: This is the power source of the project. A battery's main function is to store and provide electrical energy by transforming chemical energy into electrical energy. It allows for portable power in devices, serves as backup power during outages, and helps stabilize energy supply in renewable systems. Batteries are crucial for powering various applications in everyday life and industry.



Temperature: A temperature sensor detects temperature and converts it into a readable format, typically in degrees Celsius or Fahrenheit. Its key functions include accurately measuring temperature changes, monitoring environments (such as weather stations and HVAC systems), ensuring safe operation in industrial settings, triggering alarms to prevent overheating, and gathering data for research and analysis. In summary, temperature sensors are vital for maintaining safety and efficiency across a wide range of applications.



Methodology:

The methodology for a liquid monitoring system project involves several key steps. First, sensors are selected based on the specific liquid parameters to be measured, such as pH, temperature, or chemical concentration. These sensors are then installed in the monitoring environment, whether it's a water body, industrial process, or other application. Data collection is carried out in real-time, with the sensors transmitting data wirelessly to a central control system or cloud platform. Advanced software tools analyze this data to provide actionable insights, detect anomalies, or trigger alarms in case of threshold breaches. Regular calibration and maintenance are performed to ensure sensor accuracy, and the entire system's performance is monitored continuously. Integration with IoT and machine learning technologies is often employed to enhance data processing and predictive maintenance.

Result:

TDS Sensor:

Tap Water: 500 ppm, EC: 1 mS/cm

Distilled Water: 10 ppm, EC: 0.02 mS/cm

River Water: 1200 ppm, EC: 2.4 mS/cm

Temperature Sensor:

Room Temperature Water: 25°C

Heated Water: 60°C

Cold Water: 5°C

pH Sensor:

Tap Water: pH 7

Lemon Juice: pH 2

Soap Solution: pH 10

Turbidity Sensor:

Clear Water: 10 NTU

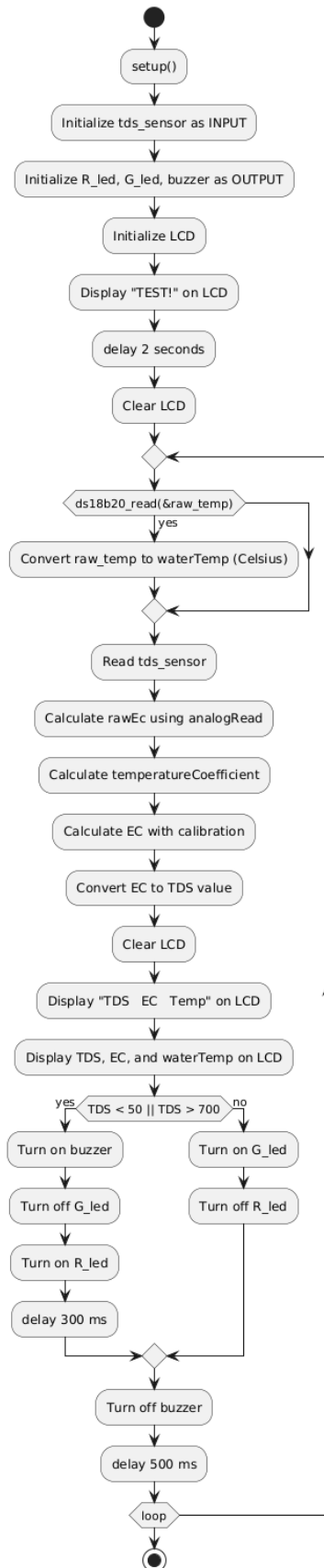
Muddy Water: 300 NTU

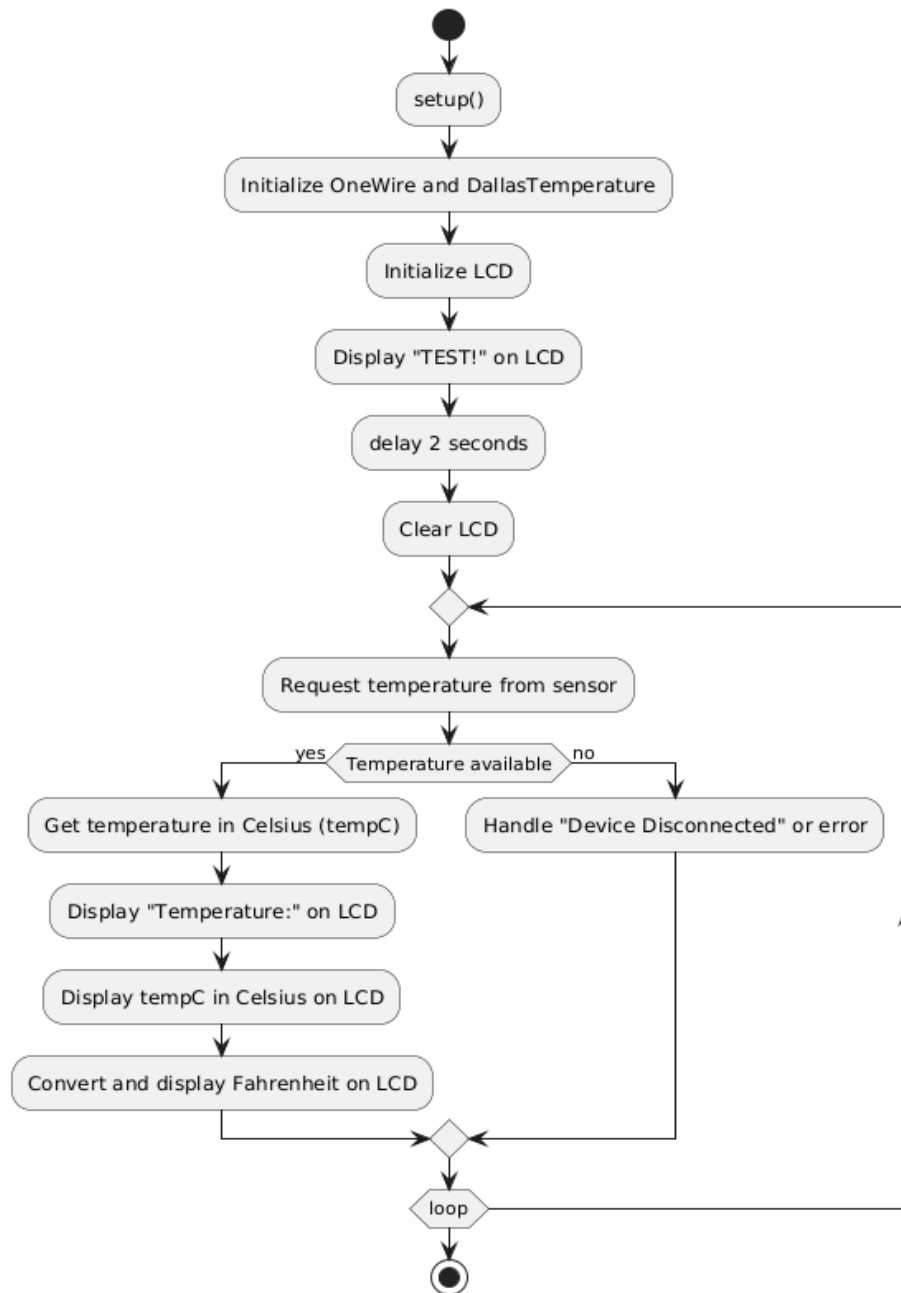
Industrial Effluent: 700 NTU

Code:

Here is the flow chart for our code. Code is given in below also.







► Code link:

<https://drive.google.com/drive/folders/1yX2y4QSORoCIECM1pEAmUtibaJMF1Vlw>

Discussion:

A liquid monitoring system is essential across various industries for real-time tracking of liquid parameters, ensuring operational efficiency, safety, and regulatory compliance. It plays a crucial role in water quality monitoring for drinking water, wastewater management, agriculture, and environmental protection, as well as in industrial applications like chemical processing, pharmaceuticals, food and beverage production, and oil and gas. Key aspects of system performance include sensor accuracy, response time, data transmission, and power efficiency. However, challenges such as sensor calibration, high implementation costs, data management, and environmental interference remain significant. Future improvements will focus on integrating AI and machine learning for better data analysis, miniaturizing sensors, enhancing wireless connectivity, and improving energy efficiency, ultimately making systems more reliable and easier to maintain in remote or industrial environments.

Challenges:

The challenges in implementing liquid monitoring systems include the need for frequent sensor calibration to maintain accuracy, particularly in harsh environments where sensor fouling or contamination can occur. High costs for installation and maintenance can be a barrier, especially for smaller businesses or municipalities. Managing the large amounts of data generated by continuous monitoring requires robust data processing, storage, and analysis systems, which can be resource-intensive. Environmental factors, such as interference or contamination, can affect sensor performance, leading to the need for frequent cleaning or recalibration. Additionally, energy requirements, especially for remote or off-grid systems, pose challenges in ensuring long-term, reliable operation without frequent maintenance.

Limitation:

Limitations of liquid monitoring systems include their dependency on consistent maintenance and calibration to ensure accurate readings over time. Many systems face challenges in extreme environments where sensor durability may be compromised, leading to reduced lifespan or malfunction. The high initial cost and ongoing operational expenses can limit widespread adoption, particularly in small-scale or resource-constrained settings. Data management can also be a limitation, as the large volumes of data generated need advanced analytics and storage solutions, which can strain existing infrastructure. Additionally, current systems may struggle with connectivity in remote locations, and energy consumption remains a concern, especially for systems that cannot easily access reliable power sources.